

STATE UNIVERSITY OF ZANZIBAR (SUZA)

School of Computing, Communication and Media Studies (SCCMS)

Department of Computer Science & Information Technology

Web Technologies

SOFTWARE REQUIREMENTS SPECIFICATION

PROJECT TITLE: University Innovation Connect (UIC): A Web-Based Collaborative Platform for Student Idea Sharing and Lecturer Feedback at SUZA

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the **University Innovation Connect (UIC)** system. This document describes what the system shall do and serves as a reference for the design, development, testing, and evaluation of the system.

This SRS document is intended for the project supervisor, developer, examiners, and other stakeholders. It serves as a common reference to ensure that all stakeholders have a clear and shared understanding of the system requirements before implementation begins.

1.2 Project Scope

The **University Innovation Connect (UIC)** is a web-based platform developed to support innovation and collaboration among students and lecturers at the State University of Zanzibar (SUZA). The system shall provide a centralized platform where students can register, log in, submit innovative ideas, view ideas posted by others, comment on ideas, and vote for the best ideas. Lecturers shall be able to review submitted ideas and provide feedback to guide students in improving their innovations.

The system is intended to improve communication, preserve student ideas, and promote collaboration within the university. It shall be developed using React for the front-end, Spring Boot for the back-end, and PostgreSQL for data storage.

The system shall not include a mobile application, artificial intelligence features, real-time chat, video conferencing, or integration with other university information systems.

1.3 Definitions, Acronyms, and Abbreviations

Term / Acronym	Definition
SRS	Software Requirements Specification
UI	User Interface
SUZA	State University of Zanzibar
API	Application Programming Interface
ERD	Entity Relationship Diagram
FR	Functional Requirement
NFR	Non-Functional Requirement
DBMS	Database Management System
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
UML	Unified Modeling Language
REST API	Representational State Transfer Application Programming Interface

1.4 References

- [1] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 8th ed., McGraw-Hill, 2019.
- [2] M. Fowler, *Patterns of Enterprise Application Architecture*, Addison-Wesley, 2002.
- [3] React Documentation, "React – A JavaScript library for building user interfaces," <https://react.dev/>
- [4] Spring Boot Documentation, "Spring Boot Reference Guide," <https://spring.io/projects/spring-boot>

1.5 Overview of the Document

This Software Requirements Specification (SRS) is organized into six main sections.

- **Section 1** introduces the purpose, scope, definitions, references, and overall organization of the document.
- **Section 2** provides an overall description of the system, including its functions, users, operating environment, constraints, and assumptions.
- **Section 3** describes the external interface requirements, including user, hardware, software, and communication interfaces.
- **Section 4** specifies the functional requirements of the system, organized by individual features.
- **Section 5** defines the non-functional requirements, including performance, security, usability, reliability, maintainability, and scalability.
- **Section 6** presents additional requirements and appendices, including data requirements, analysis models, and the requirements traceability matrix.

2. Overall Description

2.1 Product Perspective

The **University Innovation Connect (UIC)** is a web-based application developed to support idea sharing and collaboration between students and lecturers at the State University of Zanzibar (SUZA). The system provides a centralized platform where users can manage innovation-related activities through a web browser.

The system follows a three-tier architecture consisting of the presentation layer (React), application layer (Spring Boot), and database layer (PostgreSQL). Users interact with the system through the web interface, while the application processes requests and stores data in the database.

The UIC system operates independently and does not integrate with other university information systems. It requires an internet connection for users to access its services.

2.2 Product Functions

The University Innovation Connect (UIC) system shall provide the following major functions:

- User registration and login.
- User profile management.
- Posting and managing innovative ideas.
- Viewing and searching for ideas.
- Commenting on ideas.
- Voting or liking ideas.
- Lecturer feedback on student ideas.
- Administrator management of users and system content.
- Logout from the system

2.3 User Classes and Characteristics

User Class	Characteristics	Privileges
Administrator	Has knowledge of the system and manages its overall operation.	Manage user accounts, manage ideas and comments, monitor system activities, and maintain the system.
Lecturer	Registered academic staff who use the system to review and guide student innovations.	Log in, view ideas, comment on ideas, provide feedback, and manage their profile.
Student	Registered student who uses the system to share and discuss innovative ideas.	Register, log in, create and manage ideas, view ideas, comment, vote, and manage their profile.

2.4 Operating Environment

The University Innovation Connect (UIC) system will operate in the following environment:

- **Client:** Modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari on desktop and laptop computers.
- **Server:** Spring Boot application running on a Windows or Linux operating system with Java Development Kit (JDK 17 or later).
- **Database:** PostgreSQL.
- **Network:** Internet connection using HTTP/HTTPS protocols.
- **Development Environment:** Visual Studio Code, Git, GitHub, and Postman.

2.5 Design and Implementation Constraints

The development of the University Innovation Connect (UIC) system is subject to the following constraints:

- The system shall be developed using **React.js** for the front-end, **Spring Boot** for the back-end, and **PostgreSQL** as the database.
- The system shall be developed within the project timeline set by the State University of Zanzibar (SUZA).
- The system shall be accessible through modern web browsers.
- The system shall require an internet connection for users to access its services.
- The project shall use Git and GitHub for version control and source code management.

2.6 Assumptions and Dependencies

The development and operation of the University Innovation Connect (UIC) system are based on the following assumptions and dependencies:

- Users shall have access to a stable internet connection to use the system.
- Users shall access the system using a modern web browser.
- The server shall have Java, Spring Boot, and PostgreSQL installed and configured correctly.
- Users are expected to have basic computer and web browsing skills.
- The system depends on the availability of GitHub for version control during development.

3. External Interface Requirements

3.1 User Interfaces

The University Innovation Connect (UIC) system shall provide a simple, user-friendly, and responsive web interface that is accessible through modern web browsers. The interface shall ensure easy navigation and consistency across all pages.

The main user interfaces of the system shall include:

- **Home Page:** Displays general information about the system and navigation links.
- **Registration Page:** Allows new users to create an account.
- **Login Page:** Allows registered users to securely access the system.
- **Dashboard:** Displays system information and provides access to features based on the user's role.
- **Idea Management Page:** Allows students to create, edit, view, and manage innovative ideas.
- **Idea Details Page:** Displays complete information about an idea, including comments and votes.
- **Profile Page:** Allows users to view and update their personal information.
- **Administration Page:** Allows administrators to manage users, ideas, and system content.

The interface shall use clear menus, readable fonts, consistent colors, and responsive layouts to provide a good user experience on different screen sizes.

3.2 Hardware Interfaces

The University Innovation Connect (UIC) system does not require any specialized hardware interfaces. Users shall access the system using standard computing devices such as desktop computers or laptops with a keyboard, mouse, and an internet connection.

The system shall not interact directly with external hardware devices such as printers, scanners, sensors, or biometric devices.

3.3 Software Interfaces

The University Innovation Connect (UIC) system shall interact with the following software components:

Interface	Purpose	Data / Protocol
React Front-end	Provides the user interface for students, lecturers, and administrators	HTTP/HTTPS
Spring Boot REST API	Processes user requests and business logic	JSON over HTTP/HTTPS
PostgreSQL Database	Stores and retrieves system data such as users, ideas, comments, and votes	SQL
GitHub	Stores and manages the project source code during development	Git
Web Browser	Allows users to access the system	HTTP/HTTPS

3.4 Communication Interfaces

The University Innovation Connect (UIC) system shall communicate over the internet using the **HTTP** and **HTTPS** protocols.

The communication requirements are as follows:

- The system shall use **HTTPS** to ensure secure communication between users and the server.
- The front-end shall communicate with the back-end through **RESTful APIs** using **JSON** data format.
- The system shall require a stable internet connection for users to access its services.
- The server shall process user requests and return responses through standard web communication protocols.

4. System Features (Functional Requirements)

4.1 Feature: User Registration and Login

4.1.1 Description

This feature allows students and lecturers to create an account and securely log into the University Innovation Connect (UIC) system. It ensures that only registered users can access the system and its services according to their assigned roles.

4.1.2 Functional Requirements

Field	Description
Requirement ID	FR-01
Title	User Registration and Login
Description	The system shall allow students and lecturers to register, log in securely, and access the system according to their assigned roles.
Priority	High
Actor(s)	Student, Lecturer
Pre-condition	The user has access to the system through a web browser.
Trigger	The user selects Register or Login from the home page.
Main flow	1. The user opens the registration or login page. 2. The user enters the required information. 3. The system validates the entered information. 4. If the information is valid, the system creates the account or authenticates the user. 5. The system redirects the user to the appropriate dashboard.
Alternate / Error flow	If required information is missing or incorrect, the system shall display an appropriate error message and allow the user to try again.
Post-condition	The user account is created successfully or the user is logged into the system.

4.2 Feature: Idea Management

4.2.1 Description

This feature allows students to submit, view, update, and delete their innovative ideas. It provides a centralized platform for sharing ideas with other users and receiving feedback from lecturers.

4.2.2 Functional Requirements

Field	Description
Requirement ID	FR-02
Title	Idea Management
Description	The system shall allow students to create, view, edit, and delete their innovative ideas.
Priority	High
Actor(s)	Student
Pre-condition	The student is logged into the system.
Trigger	The student selects the Post Idea or Manage Ideas option.
Main Flow	1. The student opens the idea management page. 2. The student enters or updates the idea details. 3. The system validates the information. 4. The system saves or updates the idea in the database. 5. The system displays a confirmation message.
Alternate / Error Flow	If required information is missing or invalid, the system shall display an error message and allow the student to correct the information.
Post-condition	The idea is successfully created, updated, viewed, or deleted.

4.3 Feature: Comments and Lecturer Feedback

4.3.1 Description

This feature allows students to comment on ideas and enables lecturers to provide feedback and guidance. It promotes collaboration and helps improve the quality of innovative ideas.

4.3.2 Functional Requirements

Field	Description
Requirement ID	FR-03
Title	Comments and Lecturer Feedback
Description	The system shall allow students to comment on ideas and lecturers to provide feedback on submitted ideas.
Priority	High
Actor(s)	Student, Lecturer
Pre-condition	The user is logged into the system and an idea exists.
Trigger	The user selects an idea and submits a comment or feedback.
Main Flow	1. The user opens the idea details page. 2. The user enters a comment or feedback. 3. The system validates the input. 4. The system saves the comment or feedback. 5. The system displays the new comment or feedback on the idea page.

Field	Description
Alternate / Error Flow	If the comment or feedback is empty or invalid, the system shall display an error message and request the user to enter valid information.
Post-condition	The comment or feedback is successfully stored and displayed.

4.4 Feature: Idea Voting

4.4.1 Description

This feature allows registered users to vote for innovative ideas. Voting helps identify popular ideas and encourages students to improve their innovations based on community support.

4.4.2 Functional Requirements

Field	Description
Requirement ID	FR-04
Title	Idea Voting
Description	The system shall allow registered users to vote for innovative ideas.
Priority	High
Actor(s)	Student, Lecturer
Pre-condition	The user is logged into the system and the selected idea exists.
Trigger	The user clicks the Vote button on an idea.
Main Flow	1. The user views an idea. 2. The user clicks the Vote button. 3. The system verifies the user's request. 4. The system records the vote. 5. The updated vote count is displayed.
Alternate / Error Flow	If the user has already voted for the idea or the idea does not exist, the system shall display an appropriate error message.
Post-condition	The vote is successfully recorded, and the vote count is updated.

4.5 Feature: User Profile Management

4.5.1 Description

This feature allows registered users to view and update their personal information. It enables students and lecturers to keep their profile details accurate and up to date.

4.5.2 Functional Requirements

Field	Description
Requirement ID	FR-05
Title	User Profile Management
Description	The system shall allow registered users to view and update their profile information.
Priority	Medium
Actor(s)	Student, Lecturer
Pre-condition	The user is logged into the system.
Trigger	The user selects the Profile option from the dashboard.
Main Flow	1. The user opens the profile page. 2. The system displays the user's current information. 3. The user updates the required information. 4. The system validates the entered information. 5. The system saves the updated profile and displays a confirmation message.
Alternate / Error Flow	If invalid or incomplete information is entered, the system shall display an error message and request the user to correct it.
Post-condition	The user's profile information is successfully updated.

4.6 Feature: Administrator Management

4.6.1 Description

This feature allows the administrator to manage the overall system, including user accounts, ideas, and comments. It ensures proper control and maintenance of the University Innovation Connect (UIC) system.

4.6.2 Functional Requirements

Field	Description
Requirement ID	FR-06
Title	Administrator Management
Description	The system shall allow the administrator to manage users, ideas, comments, and system content.
Priority	High
Actor(s)	Administrator
Pre-condition	The administrator is logged into the system.
Trigger	The administrator selects the admin dashboard.
Main Flow	1. The administrator logs into the system. 2. The system displays the admin dashboard. 3. The administrator selects a management function (users, ideas, comments). 4. The system displays the selected data. 5. The administrator performs actions such as view, delete, or update records. 6. The system saves changes and updates the database.
Alternate / Error Flow	If invalid actions are performed or unauthorized access is detected, the system shall deny access and display an error message.
Post-condition	The system data is updated and maintained by the administrator.

4.3 Functional Requirements Summary (Traceability)

Req. ID	Requirement (The system shall...)	Priority
FR-01	allow students and lecturers to register and log in securely	High
FR-02	allow students to create, view, edit, and delete innovative ideas	High
FR-03	allow students to comment on ideas and lecturers to provide feedback	High
FR-04	allow registered users to vote for ideas	High
FR-05	allow users to view and update their profile information	Medium
FR-06	allow the administrator to manage users, ideas, and comments	High

5. Non-Functional Requirements

5.1 Performance Requirements

- The system shall respond to user requests within **3 seconds** under normal network conditions.
- The system shall support at least **100 concurrent users** without performance degradation.
- The system shall load pages efficiently with minimal delay in standard web browsers.

5.2 Security Requirements

- The system shall authenticate all users before granting access to protected features.
- The system shall use **encrypted passwords (hashing)** for secure storage.
- The system shall implement **role-based access control (RBAC)** for Students, Lecturers, and Administrators.
- The system shall use **HTTPS** to secure data transmission.

5.3 Usability Requirements

- The system shall provide a simple and user-friendly interface.
- A new user shall be able to register and use the system without training.
- The system shall be accessible through modern web browsers on desktop devices.

5.4 Reliability and Availability

- The system shall be available at least **99% of the time** during normal operation.
- The system shall recover gracefully from unexpected errors without data loss.
- The system shall ensure data consistency in the database at all times.

5.5 Maintainability and Portability

- The system code shall follow clean coding standards and be well documented.
- The system shall be easy to update and extend with new features.
- The system shall run on any modern operating system with a web browser.

5.6 Scalability

- The system database shall support future growth in the number of users and ideas.
- The system architecture shall allow expansion without major redesign.
- The system shall handle increased traffic by scaling server resources if needed.

5.7 Non-Functional Requirements Summary

Req. ID	Quality Attribute	Requirement	Measure / Target
NFR-01	Performance	System response time	Less than 3 seconds
NFR-02	Security	Authentication & encryption	Hashed passwords + HTTPS
NFR-03	Usability	Ease of use	No training required
NFR-04	Reliability	System uptime	99% availability
NFR-05	Scalability	User capacity	Greater than 100 concurrent users

6. Other Requirements and Appendices

6.1 Data Requirements

The University Innovation Connect (UIC) system shall store and manage the following key data entities:

- **Users:** Includes student, lecturer, and administrator account information such as name, email, password, and role.
- **Ideas:** Stores information about student-submitted innovative ideas including title, description, and timestamp.
- **Comments:** Stores feedback and discussions on ideas from students and lecturers.
- **Votes:** Records user votes on ideas to determine popularity.

All data shall be stored securely in a **PostgreSQL database**, and regular backups shall be considered to prevent data loss.

6.2 Legal, Regulatory, and Ethical Requirements

- The system shall ensure user privacy by protecting personal data such as emails and passwords.
- The system shall comply with basic data protection principles by not sharing user data with unauthorized parties.
- Users shall be informed that their data is used only for academic and system functionality purposes.
- The system shall not store sensitive personal information beyond what is required for authentication and functionality.

6. 3 Requirements Traceability Matrix

Req. ID	Source / Objective	Design Element	Test Case
FR-01	User authentication requirement	Login/Register module	TC-01
FR-02	Idea management requirement	Idea controller/service	TC-02
FR-03	Feedback system requirement	Comment module	TC-03
FR-04	Voting requirement	Vote service	TC-04
FR-05	Profile management requirement	User profile module	TC-05
FR-06	Admin control requirement	Admin dashboard	TC-06
NFR-01	Performance	System optimization	TC-07
NFR-02	Security	Authentication & encryption	TC-08

Student Attestation

We confirm that the intellectual content of this document the analysis, design decisions, and judgments is our own work. I have declared all AI assistance above. I understand that I may be asked to explain and defend any part of this submission orally, and that being unable to do so may be treated as evidence of misconduct

Project Repository (Version Control)

Field	Value
Repository URL	https://github.com/KhadijaSaid07/University-Innovation-Connect
Date of first commit	June 2026
Approx. number of commits	03
Supervisor granted access?	Yes

AI Tools Used in This Submission

AI Tool / Service	Specifically used for
ChatGPT	Providing guidance on formatting, and academic writing style (including IEEE references)